

# TidyTuesday Week 37

Pranay Gundam

Saturday 12<sup>th</sup> September, 2026

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# 1 Weekly Summary

## AI-Generated Content

The summary below was written by Claude (Anthropic) from that week's regression outputs and series descriptions. It has not been reviewed or fact-checked by a human, and should be read as an automated, informal recap – not analysis.

Welcome back to the segment where we bravely fling unrelated economic data at each other and pretend to be surprised by what sticks. This week the random number gods really outdid themselves, opening on Monday with a matchup between the "Discussion About Pandemics Index for Lithuania" and U.S. federal borrowing from the public. Two things that have absolutely nothing to do with each other, and yet — brace yourselves — the raw regression came back significant, and it actually survived detrending too. So even after we strip out the shared drift, Lithuanian pandemic chatter and American public debt still hold hands a little. Is it real? Almost certainly not. Am I contractually obligated to note that this is the "rarer, more interesting case"? Yes, and I hate that.

Tuesday paired U.S. coal CO2 emissions with a medical price index for congenital anomalies, and honestly this one is the overachiever of the week. The raw regression explained about 61 percent of the variation, and instead of collapsing when detrended, it held on at a robust 56 percent with the slope actually getting steeper. I want to be clear that there is no universe in which coal smoke is setting the price of specialized medical services, but the numbers are sitting here looking me dead in the eye like they mean it. Wednesday, meanwhile, was pure comedy: gross federal debt versus business equipment leases owned by finance companies produced a raw relationship with a laughably tiny R-squared, and detrending somehow made it "more significant" while explaining almost nothing. Truly the statistical equivalent of confidently walking into the wrong room.

Thursday gave us the week's designated punching bag, and I love it dearly. Construction quits versus the work-from-home rate for women started off with an R-squared of essentially zero and a p-value of 0.86 — the raw regression basically shrugged and left. Then we detrend, and suddenly it lurches to life with a chunky, very significant relationship out of nowhere. This is a beautiful reminder that "significant after detrending" is evidence, not proof, and that random noise is perfectly capable of getting dressed up for the occasion. Friday's national wages versus manufacturing employment in Ogden, Utah was the classic trend mirage: an eye-watering 80 percent R-squared in the raw version that deflated to a much humbler 40 percent once the shared upward march of time was removed. Impressive headline, ordinary story.

We closed Saturday with Midwest shelter prices versus the price of a whole chicken in the West, which posted the flashiest raw number of the entire week — 94 percent R-squared, the kind of figure that makes people quit their jobs to write a book. And then detrending did what detrending does, dragging it down to a merely-fine 37 percent. Both things go up over time

because most prices go up over time; shocking, I know. As always, the mandatory disclaimer: these are randomly paired series with zero causal claim intended, and spurious correlation is the whole point of the exercise, not an unfortunate accident. If you took investment advice from a chicken and a shelter index this week, that is between you and your conscience.

## 2 Date: 2026-09-07

### Series ID: WPDILTU

This series is titled Discussion About Pandemics Index for Lithuania and has a frequency of Quarterly. The units are Index and the seasonal adjustment is Not Seasonally Adjusted. The observation start date is 1996-01-01 and the observation end date is 2026-04-01. The popularity of this series is 1.

### Series ID: MTSMFBP133FMS

This series is titled Means of Financing: Borrowing from the Public and has a frequency of Monthly. The units are Millions of Dollars and the seasonal adjustment is Not Seasonally Adjusted. The observation start date is 2010-01-01 and the observation end date is 2026-07-01. The popularity of this series is 14.

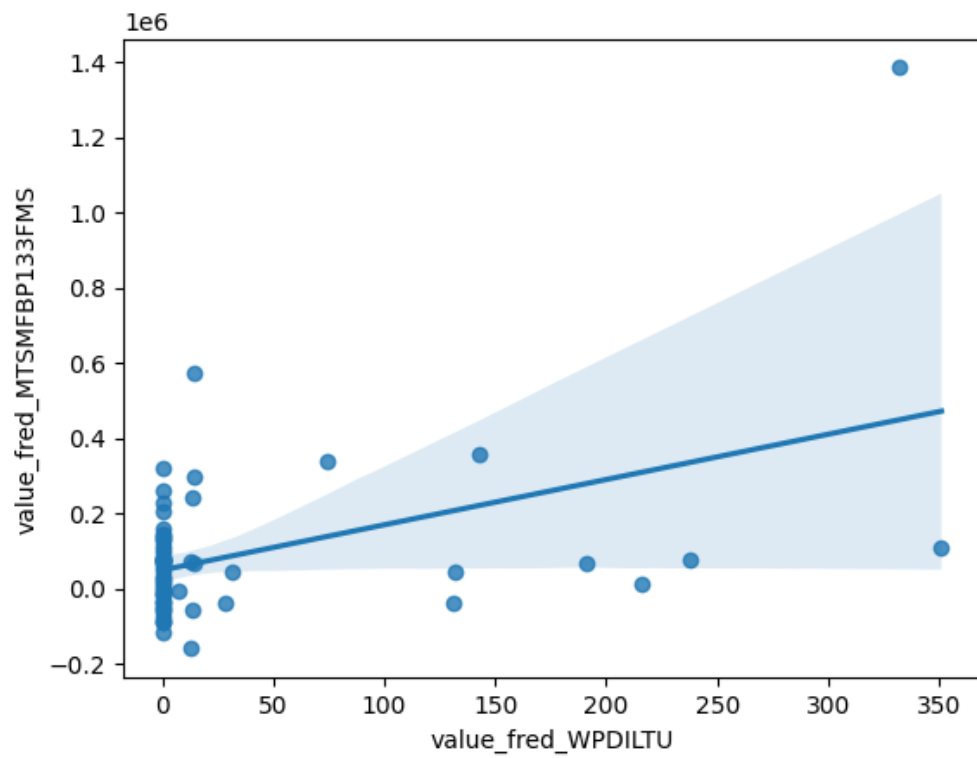
## 2.1 Raw Regression

Regresses the two series against each other directly. A significant result here can come just as easily from both series sharing a long-run trend as from any real relationship.

|                    |                          |                     |          |       |          |          |
|--------------------|--------------------------|---------------------|----------|-------|----------|----------|
| Dep. Variable:     | value_fred_MTSMFBP133FMS | R-squared:          | 0.199    |       |          |          |
| Model:             | OLS                      | Adj. R-squared:     | 0.186    |       |          |          |
| Method:            | Least Squares            | F-statistic:        | 15.87    |       |          |          |
| Date:              | Mon, 07 Sep 2026         | Prob (F-statistic): | 0.000176 |       |          |          |
| Time:              | 16:43:46                 | Log-Likelihood:     | -893.46  |       |          |          |
| No. Observations:  | 66                       | AIC:                | 1791.    |       |          |          |
| Df Residuals:      | 64                       | BIC:                | 1795.    |       |          |          |
| Df Model:          | 1                        |                     |          |       |          |          |
| Covariance Type:   | nonrobust                |                     |          |       |          |          |
|                    | coef                     | std err             | t        | P>  t | [0.025   | 0.975]   |
| const              | 4.886e+04                | 2.46e+04            | 1.985    | 0.051 | -321.644 | 9.8e+04  |
| value_fred_WPDILTU | 1205.6312                | 302.613             | 3.984    | 0.000 | 601.092  | 1810.170 |
| Omnibus:           | 51.230                   | Durbin-Watson:      | 2.093    |       |          |          |
| Prob(Omnibus):     | 0.000                    | Jarque-Bera (JB):   | 275.412  |       |          |          |
| Skew:              | 2.139                    | Prob(JB):           | 1.57e-60 |       |          |          |
| Kurtosis:          | 12.047                   | Cond. No.           | 87.5     |       |          |          |

Notes:

[1] Standard Errors assume that the covariance matrix of the errors is correctly specified.



**Figure 1:** Raw Regression Plot for 2026-09-07

## 2.2 Detrended Regression

Regresses the residuals of each series after removing its own linear time trend, so a shared trend can no longer drive the result on its own. A relationship that stays significant here is better evidence of a genuine link between the two series.

|                          |                                    |                            |          |
|--------------------------|------------------------------------|----------------------------|----------|
| <b>Dep. Variable:</b>    | value_fred_MTSMFBP133FMS_detrended | <b>R-squared:</b>          | 0.174    |
| <b>Model:</b>            | OLS                                | <b>Adj. R-squared:</b>     | 0.161    |
| <b>Method:</b>           | Least Squares                      | <b>F-statistic:</b>        | 13.48    |
| <b>Date:</b>             | Mon, 07 Sep 2026                   | <b>Prob (F-statistic):</b> | 0.000493 |
| <b>Time:</b>             | 16:43:46                           | <b>Log-Likelihood:</b>     | -893.06  |
| <b>No. Observations:</b> | 66                                 | <b>AIC:</b>                | 1790.    |
| <b>Df Residuals:</b>     | 64                                 | <b>BIC:</b>                | 1794.    |
| <b>Df Model:</b>         | 1                                  |                            |          |
| <b>Covariance Type:</b>  | nonrobust                          |                            |          |

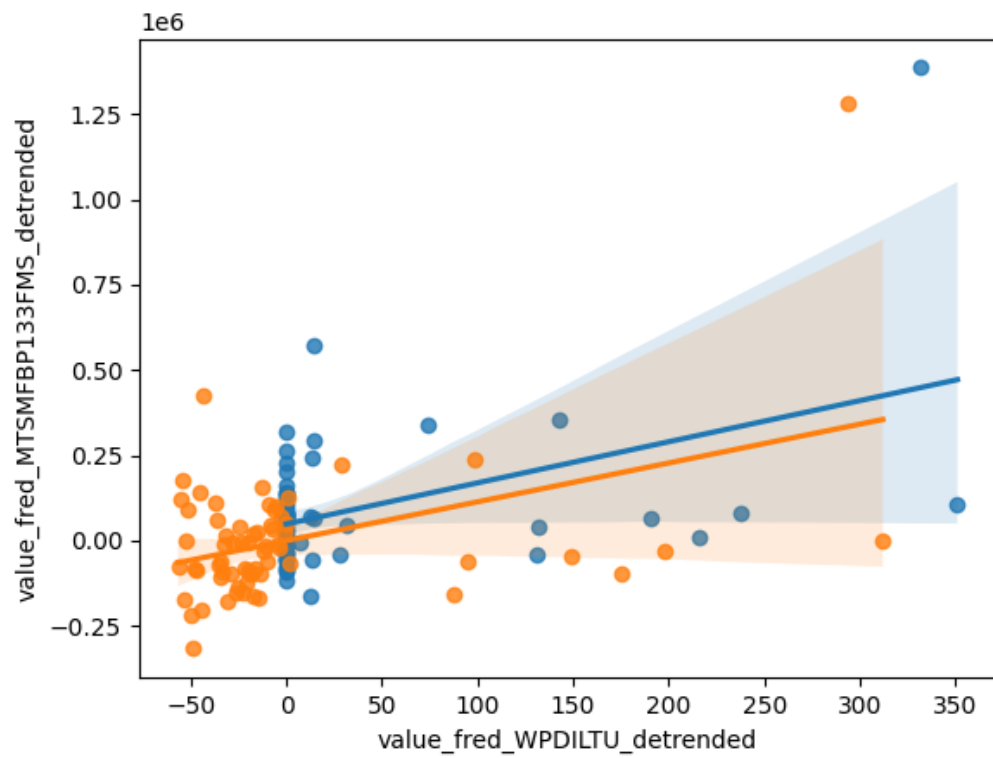
|                                     | <b>coef</b> | <b>std err</b> | <b>t</b> | <b>P&gt;  t </b> | <b>[0.025</b> | <b>0.975]</b> |
|-------------------------------------|-------------|----------------|----------|------------------|---------------|---------------|
| <b>const</b>                        | 5.051e-09   | 2.28e+04       | 2.22e-13 | 1.000            | -4.55e+04     | 4.55e+04      |
| <b>value_fred_WPDILTU_detrended</b> | 1138.7087   | 310.105        | 3.672    | 0.000            | 519.204       | 1758.214      |

|                       |        |                          |          |
|-----------------------|--------|--------------------------|----------|
| <b>Omnibus:</b>       | 52.012 | <b>Durbin-Watson:</b>    | 2.129    |
| <b>Prob(Omnibus):</b> | 0.000  | <b>Jarque-Bera (JB):</b> | 302.597  |
| <b>Skew:</b>          | 2.138  | <b>Prob(JB):</b>         | 1.96e-66 |
| <b>Kurtosis:</b>      | 12.579 | <b>Cond. No.</b>         | 73.4     |

Notes:

[1] Standard Errors assume that the covariance matrix of the errors is correctly specified.



**Figure 2:** Detrended Regression Plot for 2026-09-07

### 3 Date: 2026-09-08

#### Series ID: EMISSCO2TOTVTTTCOUSA

This series is titled Total Carbon Dioxide Emissions From All Sectors, Coal for United States and has a frequency of Annual. The units are Million Metric Tons CO2 and the seasonal adjustment is Not Seasonally Adjusted. The observation start date is 1970-01-01 and the observation end date is 2021-01-01. The popularity of this series is 5.

#### Series ID: COGANOPIMEPS

This series is titled Medical Services Expenditures by Disease: Other: Congenital Anomalies Price Index, MEPS Account Basis and has a frequency of Annual. The units are Index 2009=100 and the seasonal adjustment is Not Seasonally Adjusted. The observation start date is 2000-01-01 and the observation end date is 2020-01-01. The popularity of this series is 1.

### 3.1 Raw Regression

Regresses the two series against each other directly. A significant result here can come just as easily from both series sharing a long-run trend as from any real relationship.

|                          |                         |                            |          |
|--------------------------|-------------------------|----------------------------|----------|
| <b>Dep. Variable:</b>    | value_fred_COGANOPIMEPS | <b>R-squared:</b>          | 0.606    |
| <b>Model:</b>            | OLS                     | <b>Adj. R-squared:</b>     | 0.585    |
| <b>Method:</b>           | Least Squares           | <b>F-statistic:</b>        | 29.22    |
| <b>Date:</b>             | Tue, 08 Sep 2026        | <b>Prob (F-statistic):</b> | 3.24e-05 |
| <b>Time:</b>             | 15:27:44                | <b>Log-Likelihood:</b>     | -111.42  |
| <b>No. Observations:</b> | 21                      | <b>AIC:</b>                | 226.8    |
| <b>Df Residuals:</b>     | 19                      | <b>BIC:</b>                | 228.9    |
| <b>Df Model:</b>         | 1                       |                            |          |
| <b>Covariance Type:</b>  | nonrobust               |                            |          |

|                                        | coef      | std err | t      | P>  t | [0.025   | 0.975]  |
|----------------------------------------|-----------|---------|--------|-------|----------|---------|
| <b>const</b>                           | -146.4574 | 51.777  | -2.829 | 0.011 | -254.827 | -38.088 |
| <b>value_fred_EMISSCO2TOTVTTTCOUSA</b> | 0.1533    | 0.028   | 5.406  | 0.000 | 0.094    | 0.213   |

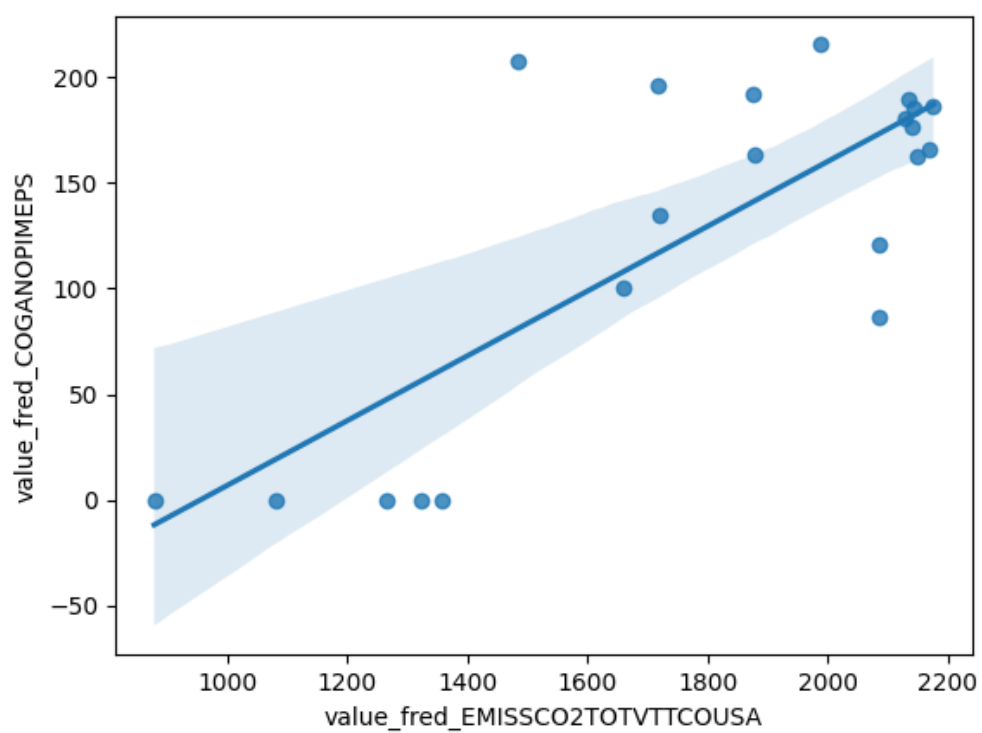
  

|                       |       |                          |          |
|-----------------------|-------|--------------------------|----------|
| <b>Omnibus:</b>       | 3.022 | <b>Durbin-Watson:</b>    | 1.167    |
| <b>Prob(Omnibus):</b> | 0.221 | <b>Jarque-Bera (JB):</b> | 1.530    |
| <b>Skew:</b>          | 0.625 | <b>Prob(JB):</b>         | 0.465    |
| <b>Kurtosis:</b>      | 3.429 | <b>Cond. No.</b>         | 8.45e+03 |

Notes:

[1] Standard Errors assume that the covariance matrix of the errors is correctly specified.

[2] The condition number is large, 8.45e+03. This might indicate that there are strong multicollinearity or other numerical problems.



**Figure 3:** Raw Regression Plot for 2026-09-08

### 3.2 Detrended Regression

Regresses the residuals of each series after removing its own linear time trend, so a shared trend can no longer drive the result on its own. A relationship that stays significant here is better evidence of a genuine link between the two series.

|                          |                                   |                            |          |
|--------------------------|-----------------------------------|----------------------------|----------|
| <b>Dep. Variable:</b>    | value_fred_COGANOPIMEPS_detrended | <b>R-squared:</b>          | 0.559    |
| <b>Model:</b>            | OLS                               | <b>Adj. R-squared:</b>     | 0.536    |
| <b>Method:</b>           | Least Squares                     | <b>F-statistic:</b>        | 24.09    |
| <b>Date:</b>             | Tue, 08 Sep 2026                  | <b>Prob (F-statistic):</b> | 9.76e-05 |
| <b>Time:</b>             | 15:27:44                          | <b>Log-Likelihood:</b>     | -108.14  |
| <b>No. Observations:</b> | 21                                | <b>AIC:</b>                | 220.3    |
| <b>Df Residuals:</b>     | 19                                | <b>BIC:</b>                | 222.4    |
| <b>Df Model:</b>         | 1                                 |                            |          |
| <b>Covariance Type:</b>  | nonrobust                         |                            |          |

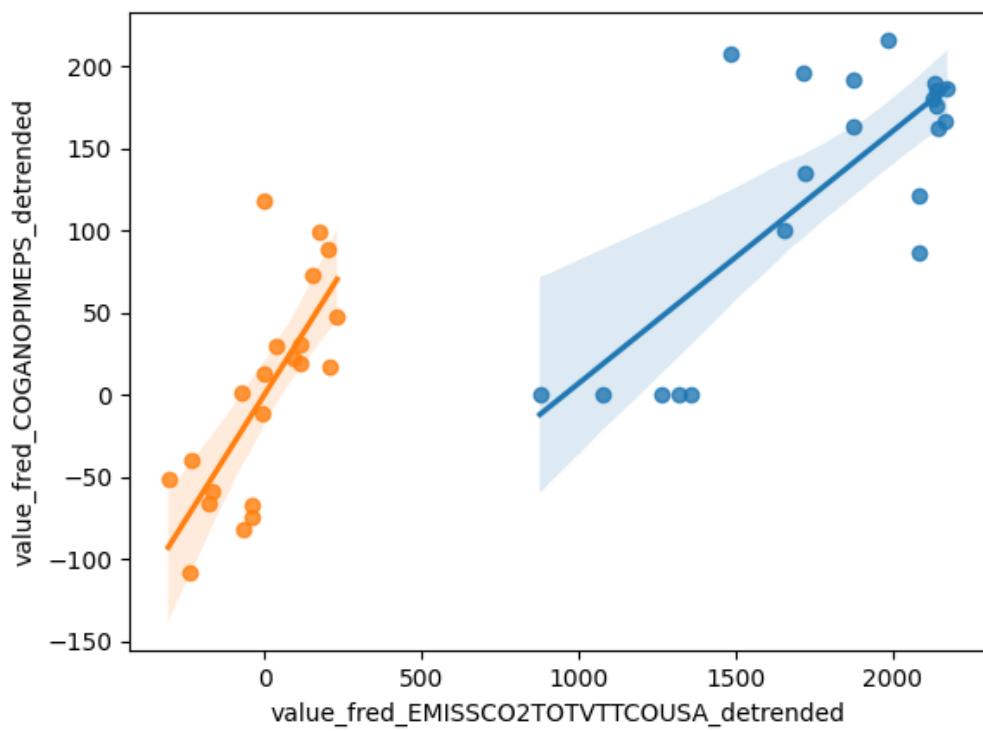
|                                                 | coef       | std err | t         | P>  t | [0.025  | 0.975] |
|-------------------------------------------------|------------|---------|-----------|-------|---------|--------|
| <b>const</b>                                    | -6.734e-12 | 9.567   | -7.04e-13 | 1.000 | -20.023 | 20.023 |
| <b>value_fred_EMISSCO2TOTVTTCOUSA_detrended</b> | 0.3032     | 0.062   | 4.908     | 0.000 | 0.174   | 0.432  |

|                       |       |                          |       |
|-----------------------|-------|--------------------------|-------|
| <b>Omnibus:</b>       | 4.692 | <b>Durbin-Watson:</b>    | 1.541 |
| <b>Prob(Omnibus):</b> | 0.096 | <b>Jarque-Bera (JB):</b> | 2.578 |
| <b>Skew:</b>          | 0.726 | <b>Prob(JB):</b>         | 0.276 |
| <b>Kurtosis:</b>      | 3.915 | <b>Cond. No.</b>         | 155.  |

Notes:

[1] Standard Errors assume that the covariance matrix of the errors is correctly specified.



**Figure 4:** Detrended Regression Plot for 2026-09-08

## 4 Date: 2026-09-09

### Series ID: MVGFD027MNFRBDAL

This series is titled Market Value of Gross Federal Debt and has a frequency of Monthly. The units are Billions of Dollars and the seasonal adjustment is Not Seasonally Adjusted. The observation start date is 1942-01-01 and the observation end date is 2026-07-01. The popularity of this series is 37.

### Series ID: DTBTERNM

This series is titled Business Equipment Leases Owned and Securitized by Finance Companies, Level and has a frequency of Monthly. The units are Millions of Dollars and the seasonal adjustment is Not Seasonally Adjusted. The observation start date is 1990-06-01 and the observation end date is 2026-06-01. The popularity of this series is 1.

## 4.1 Raw Regression

Regresses the two series against each other directly. A significant result here can come just as easily from both series sharing a long-run trend as from any real relationship.

|                          |                     |                            |           |
|--------------------------|---------------------|----------------------------|-----------|
| <b>Dep. Variable:</b>    | value_fred_DTBTERNM | <b>R-squared:</b>          | 0.095     |
| <b>Model:</b>            | OLS                 | <b>Adj. R-squared:</b>     | 0.093     |
| <b>Method:</b>           | Least Squares       | <b>F-statistic:</b>        | 45.33     |
| <b>Date:</b>             | Wed, 09 Sep 2026    | <b>Prob (F-statistic):</b> | 5.34e-11  |
| <b>Time:</b>             | 15:24:33            | <b>Log-Likelihood:</b>     | -5318.4   |
| <b>No. Observations:</b> | 433                 | <b>AIC:</b>                | 1.064e+04 |
| <b>Df Residuals:</b>     | 431                 | <b>BIC:</b>                | 1.065e+04 |
| <b>Df Model:</b>         | 1                   |                            |           |
| <b>Covariance Type:</b>  | nonrobust           |                            |           |

|                                    | coef      | std err  | t      | P>  t | [0.025   | 0.975]   |
|------------------------------------|-----------|----------|--------|-------|----------|----------|
| <b>const</b>                       | 1.633e+05 | 4472.957 | 36.517 | 0.000 | 1.55e+05 | 1.72e+05 |
| <b>value_fred_MVGFD027MNFRBDAL</b> | -1.7290   | 0.257    | -6.732 | 0.000 | -2.234   | -1.224   |

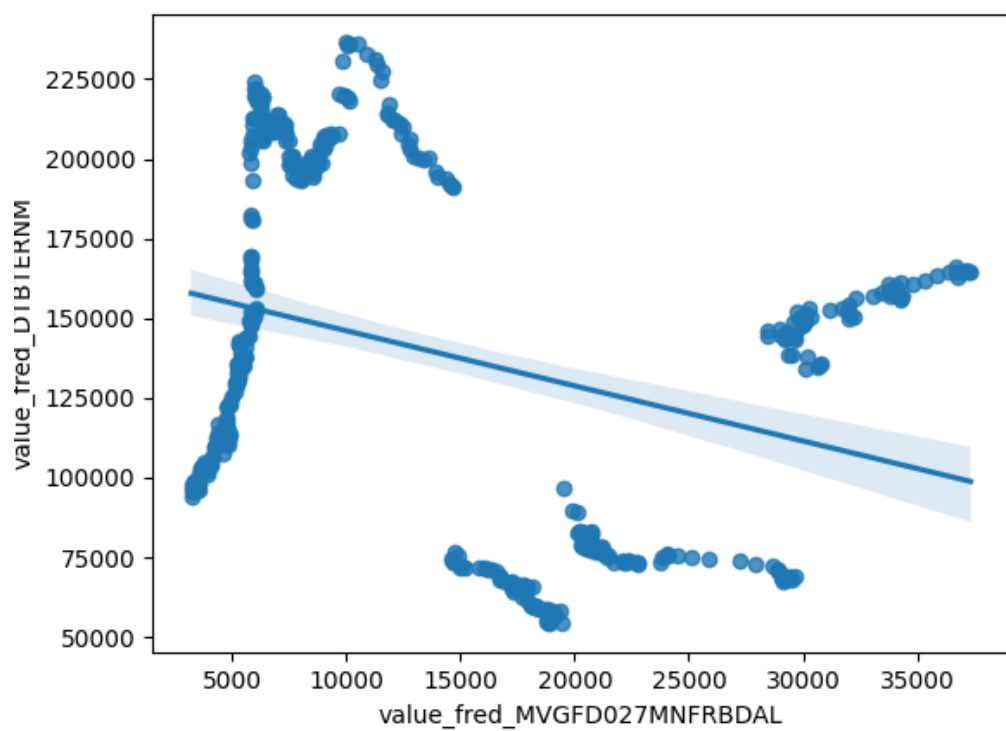
  

|                       |          |                          |          |
|-----------------------|----------|--------------------------|----------|
| <b>Omnibus:</b>       | 2749.157 | <b>Durbin-Watson:</b>    | 0.021    |
| <b>Prob(Omnibus):</b> | 0.000    | <b>Jarque-Bera (JB):</b> | 48.079   |
| <b>Skew:</b>          | 0.054    | <b>Prob(JB):</b>         | 3.63e-11 |
| <b>Kurtosis:</b>      | 1.371    | <b>Cond. No.</b>         | 3.10e+04 |

Notes:

[1] Standard Errors assume that the covariance matrix of the errors is correctly specified.

[2] The condition number is large, 3.1e+04. This might indicate that there are strong multicollinearity or other numerical problems.



**Figure 5:** Raw Regression Plot for 2026-09-09

## 4.2 Detrended Regression

Regresses the residuals of each series after removing its own linear time trend, so a shared trend can no longer drive the result on its own. A relationship that stays significant here is better evidence of a genuine link between the two series.

|                          |                               |                            |           |
|--------------------------|-------------------------------|----------------------------|-----------|
| <b>Dep. Variable:</b>    | value_fred_DTBTERNM_detrended | <b>R-squared:</b>          | 0.116     |
| <b>Model:</b>            | OLS                           | <b>Adj. R-squared:</b>     | 0.114     |
| <b>Method:</b>           | Least Squares                 | <b>F-statistic:</b>        | 56.72     |
| <b>Date:</b>             | Wed, 09 Sep 2026              | <b>Prob (F-statistic):</b> | 2.97e-13  |
| <b>Time:</b>             | 15:24:33                      | <b>Log-Likelihood:</b>     | -5302.8   |
| <b>No. Observations:</b> | 433                           | <b>AIC:</b>                | 1.061e+04 |
| <b>Df Residuals:</b>     | 431                           | <b>BIC:</b>                | 1.062e+04 |
| <b>Df Model:</b>         | 1                             |                            |           |
| <b>Covariance Type:</b>  | nonrobust                     |                            |           |

|                                              | coef      | std err  | t        | P> t  | [0.025    | 0.975    |
|----------------------------------------------|-----------|----------|----------|-------|-----------|----------|
| <b>const</b>                                 | 9.044e-09 | 2427.545 | 3.73e-12 | 1.000 | -4771.300 | 4771.300 |
| <b>value_fred_MVGFD027MNFRBDAL_detrended</b> | -6.1487   | 0.816    | -7.531   | 0.000 | -7.753    | -4.543   |

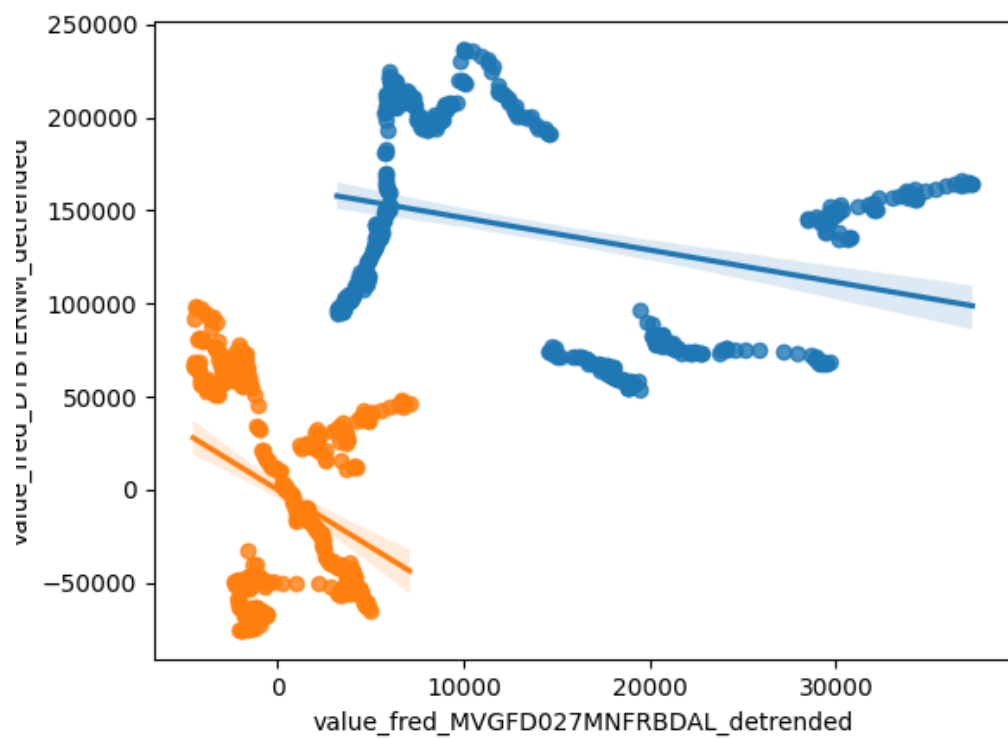
  

|                       |         |                          |          |
|-----------------------|---------|--------------------------|----------|
| <b>Omnibus:</b>       | 698.313 | <b>Durbin-Watson:</b>    | 0.023    |
| <b>Prob(Omnibus):</b> | 0.000   | <b>Jarque-Bera (JB):</b> | 33.214   |
| <b>Skew:</b>          | -0.174  | <b>Prob(JB):</b>         | 6.13e-08 |
| <b>Kurtosis:</b>      | 1.689   | <b>Cond. No.</b>         | 2.97e+03 |

Notes:

[1] Standard Errors assume that the covariance matrix of the errors is correctly specified.

[2] The condition number is large, 2.97e+03. This might indicate that there are strong multicollinearity or other numerical problems.



**Figure 6:** Detrended Regression Plot for 2026-09-09

## 5 Date: 2026-09-10

### Series ID: JTS2300QUR

This series is titled Quits: Construction and has a frequency of Monthly. The units are Rate and the seasonal adjustment is Seasonally Adjusted. The observation start date is 2000-12-01 and the observation end date is 2026-07-01. The popularity of this series is 13.

### Series ID: WFHCOVIDFRACMATWOMEN

This series is titled Work from Home Rate: Women and has a frequency of Monthly. The units are Percent of Full Paid Working Days and the seasonal adjustment is Not Seasonally Adjusted. The observation start date is 2020-05-01 and the observation end date is 2026-08-01. The popularity of this series is 6.

### 5.1 Raw Regression

Regresses the two series against each other directly. A significant result here can come just as easily from both series sharing a long-run trend as from any real relationship.

|                          |                                 |                            |         |
|--------------------------|---------------------------------|----------------------------|---------|
| <b>Dep. Variable:</b>    | value_fred_WFHCOVIDFRACMATWOMEN | <b>R-squared:</b>          | 0.000   |
| <b>Model:</b>            | OLS                             | <b>Adj. R-squared:</b>     | -0.013  |
| <b>Method:</b>           | Least Squares                   | <b>F-statistic:</b>        | 0.03323 |
| <b>Date:</b>             | Thu, 10 Sep 2026                | <b>Prob (F-statistic):</b> | 0.856   |
| <b>Time:</b>             | 15:17:47                        | <b>Log-Likelihood:</b>     | -240.33 |
| <b>No. Observations:</b> | 74                              | <b>AIC:</b>                | 484.7   |
| <b>Df Residuals:</b>     | 72                              | <b>BIC:</b>                | 489.3   |
| <b>Df Model:</b>         | 1                               |                            |         |
| <b>Covariance Type:</b>  | nonrobust                       |                            |         |

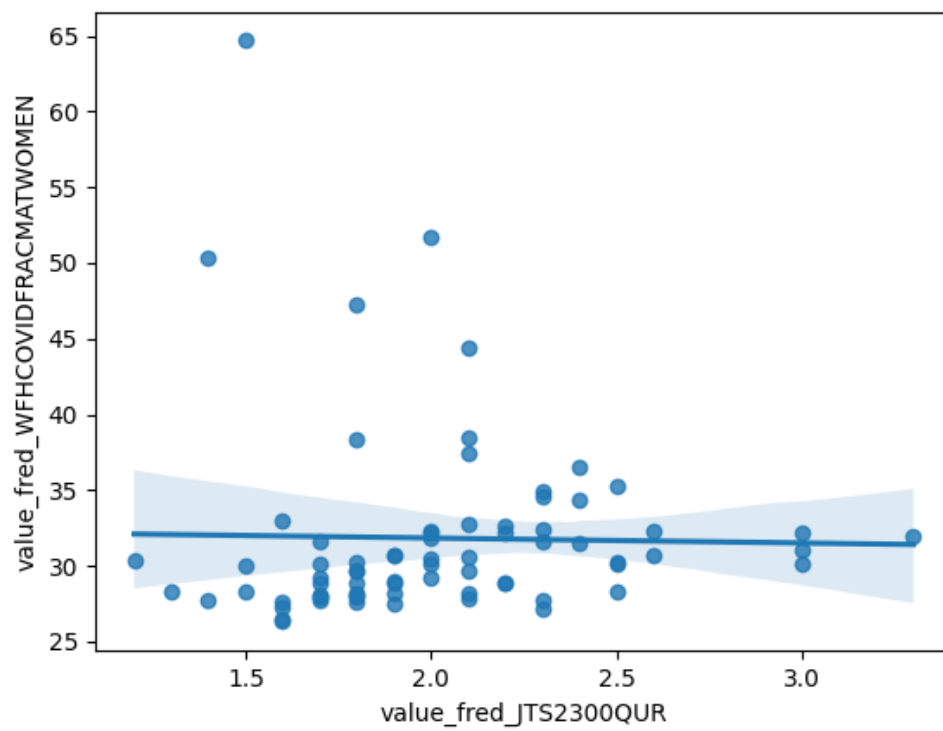
|                              | coef    | std err | t      | P>  t | [0.025 | 0.975] |
|------------------------------|---------|---------|--------|-------|--------|--------|
| <b>const</b>                 | 32.4960 | 3.704   | 8.774  | 0.000 | 25.113 | 39.879 |
| <b>value_fred_JTS2300QUR</b> | -0.3289 | 1.804   | -0.182 | 0.856 | -3.926 | 3.268  |

|                       |        |                          |          |
|-----------------------|--------|--------------------------|----------|
| <b>Omnibus:</b>       | 71.083 | <b>Durbin-Watson:</b>    | 0.194    |
| <b>Prob(Omnibus):</b> | 0.000  | <b>Jarque-Bera (JB):</b> | 449.811  |
| <b>Skew:</b>          | 2.992  | <b>Prob(JB):</b>         | 2.11e-98 |
| <b>Kurtosis:</b>      | 13.492 | <b>Cond. No.</b>         | 12.7     |

Notes:

[1] Standard Errors assume that the covariance matrix of the errors is correctly specified.



**Figure 7:** Raw Regression Plot for 2026-09-10

## 5.2 Detrended Regression

Regresses the residuals of each series after removing its own linear time trend, so a shared trend can no longer drive the result on its own. A relationship that stays significant here is better evidence of a genuine link between the two series.

|                          |                                           |                            |          |
|--------------------------|-------------------------------------------|----------------------------|----------|
| <b>Dep. Variable:</b>    | value_fred_WFHCOVIDFRACMATWOMEN_detrended | <b>R-squared:</b>          | 0.300    |
| <b>Model:</b>            | OLS                                       | <b>Adj. R-squared:</b>     | 0.290    |
| <b>Method:</b>           | Least Squares                             | <b>F-statistic:</b>        | 30.84    |
| <b>Date:</b>             | Thu, 10 Sep 2026                          | <b>Prob (F-statistic):</b> | 4.44e-07 |
| <b>Time:</b>             | 15:17:47                                  | <b>Log-Likelihood:</b>     | -205.78  |
| <b>No. Observations:</b> | 74                                        | <b>AIC:</b>                | 415.6    |
| <b>Df Residuals:</b>     | 72                                        | <b>BIC:</b>                | 420.2    |
| <b>Df Model:</b>         | 1                                         |                            |          |
| <b>Covariance Type:</b>  | nonrobust                                 |                            |          |

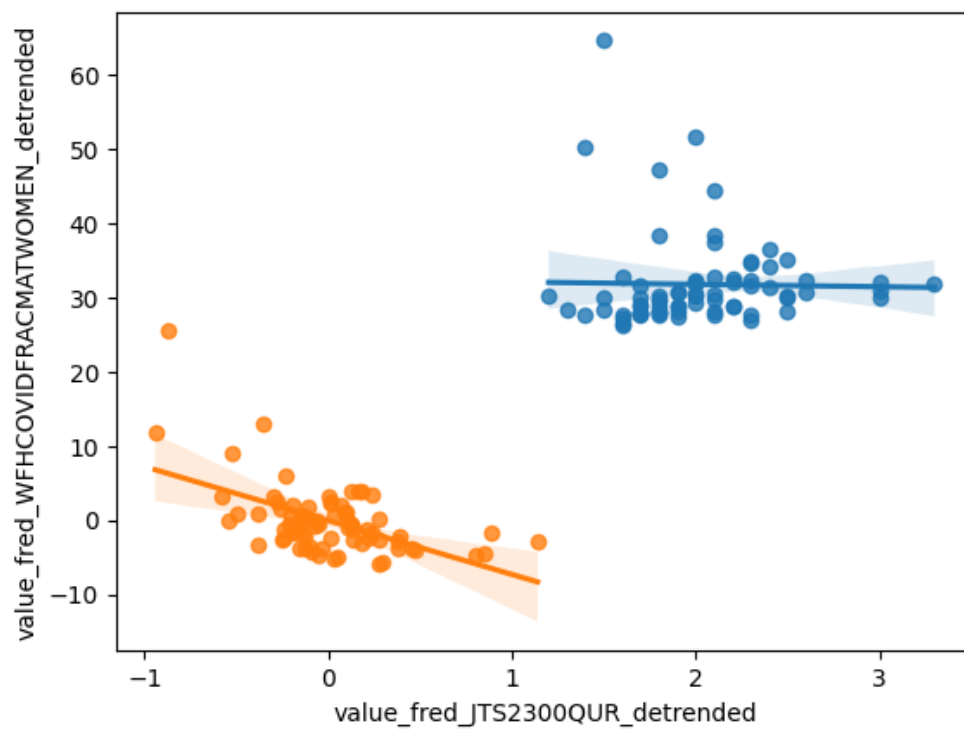
|                                        | coef       | std err | t         | P>  t | [0.025 | 0.975] |
|----------------------------------------|------------|---------|-----------|-------|--------|--------|
| <b>const</b>                           | -7.578e-12 | 0.460   | -1.65e-11 | 1.000 | -0.917 | 0.917  |
| <b>value_fred_JTS2300QUR_detrended</b> | -7.2666    | 1.309   | -5.553    | 0.000 | -9.875 | -4.658 |

|                       |        |                          |          |
|-----------------------|--------|--------------------------|----------|
| <b>Omnibus:</b>       | 45.009 | <b>Durbin-Watson:</b>    | 0.681    |
| <b>Prob(Omnibus):</b> | 0.000  | <b>Jarque-Bera (JB):</b> | 171.162  |
| <b>Skew:</b>          | 1.828  | <b>Prob(JB):</b>         | 6.80e-38 |
| <b>Kurtosis:</b>      | 9.492  | <b>Cond. No.</b>         | 2.84     |

Notes:

[1] Standard Errors assume that the covariance matrix of the errors is correctly specified.



**Figure 8:** Detrended Regression Plot for 2026-09-10

## 6 Date: 2026-09-11

### Series ID: A034RC1A027NBEA

This series is titled National income: Compensation of employees: Wages and salaries and has a frequency of Annual. The units are Billions of Dollars and the seasonal adjustment is Not Seasonally Adjusted. The observation start date is 1929-01-01 and the observation end date is 2025-01-01. The popularity of this series is 23.

### Series ID: OGDE249MFGN

This series is titled All Employees: Manufacturing in Ogden, UT (MSA) and has a frequency of Monthly. The units are Thousands of Persons and the seasonal adjustment is Not Seasonally Adjusted. The observation start date is 1990-01-01 and the observation end date is 2026-07-01. The popularity of this series is 1.

## 6.1 Raw Regression

Regresses the two series against each other directly. A significant result here can come just as easily from both series sharing a long-run trend as from any real relationship.

|                          |                        |                            |          |
|--------------------------|------------------------|----------------------------|----------|
| <b>Dep. Variable:</b>    | value_fred_OGDE249MFGN | <b>R-squared:</b>          | 0.802    |
| <b>Model:</b>            | OLS                    | <b>Adj. R-squared:</b>     | 0.797    |
| <b>Method:</b>           | Least Squares          | <b>F-statistic:</b>        | 138.0    |
| <b>Date:</b>             | Fri, 11 Sep 2026       | <b>Prob (F-statistic):</b> | 1.62e-13 |
| <b>Time:</b>             | 15:19:38               | <b>Log-Likelihood:</b>     | -76.434  |
| <b>No. Observations:</b> | 36                     | <b>AIC:</b>                | 156.9    |
| <b>Df Residuals:</b>     | 34                     | <b>BIC:</b>                | 160.0    |
| <b>Df Model:</b>         | 1                      |                            |          |
| <b>Covariance Type:</b>  | nonrobust              |                            |          |

|                                   | coef    | std err | t      | P>  t | [0.025 | 0.975] |
|-----------------------------------|---------|---------|--------|-------|--------|--------|
| <b>const</b>                      | 14.0195 | 0.885   | 15.840 | 0.000 | 12.221 | 15.818 |
| <b>value_fred_A034RC1A027NBEA</b> | 0.0015  | 0.000   | 11.748 | 0.000 | 0.001  | 0.002  |

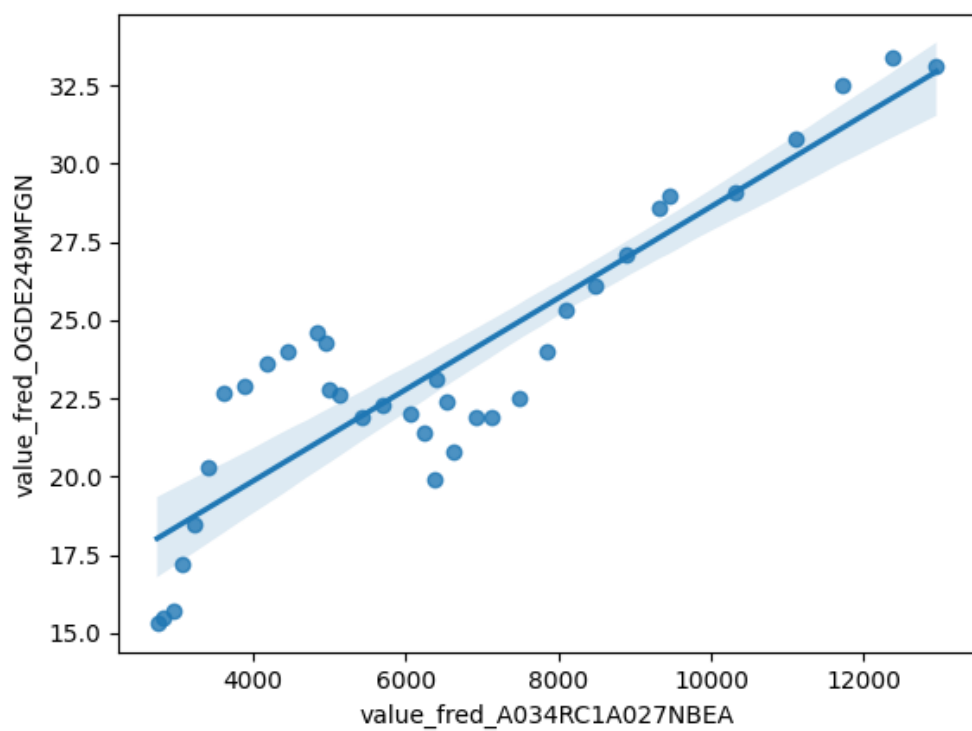
  

|                       |       |                          |          |
|-----------------------|-------|--------------------------|----------|
| <b>Omnibus:</b>       | 2.395 | <b>Durbin-Watson:</b>    | 0.177    |
| <b>Prob(Omnibus):</b> | 0.302 | <b>Jarque-Bera (JB):</b> | 1.478    |
| <b>Skew:</b>          | 0.221 | <b>Prob(JB):</b>         | 0.478    |
| <b>Kurtosis:</b>      | 2.111 | <b>Cond. No.</b>         | 1.82e+04 |

Notes:

[1] Standard Errors assume that the covariance matrix of the errors is correctly specified.

[2] The condition number is large, 1.82e+04. This might indicate that there are strong multicollinearity or other numerical problems.



**Figure 9:** Raw Regression Plot for 2026-09-11

## 6.2 Detrended Regression

Regresses the residuals of each series after removing its own linear time trend, so a shared trend can no longer drive the result on its own. A relationship that stays significant here is better evidence of a genuine link between the two series.

|                          |                                  |                            |          |
|--------------------------|----------------------------------|----------------------------|----------|
| <b>Dep. Variable:</b>    | value_fred_OGDE249MFGN_detrended | <b>R-squared:</b>          | 0.401    |
| <b>Model:</b>            | OLS                              | <b>Adj. R-squared:</b>     | 0.383    |
| <b>Method:</b>           | Least Squares                    | <b>F-statistic:</b>        | 22.75    |
| <b>Date:</b>             | Fri, 11 Sep 2026                 | <b>Prob (F-statistic):</b> | 3.40e-05 |
| <b>Time:</b>             | 15:19:38                         | <b>Log-Likelihood:</b>     | -74.459  |
| <b>No. Observations:</b> | 36                               | <b>AIC:</b>                | 152.9    |
| <b>Df Residuals:</b>     | 34                               | <b>BIC:</b>                | 156.1    |
| <b>Df Model:</b>         | 1                                |                            |          |
| <b>Covariance Type:</b>  | nonrobust                        |                            |          |

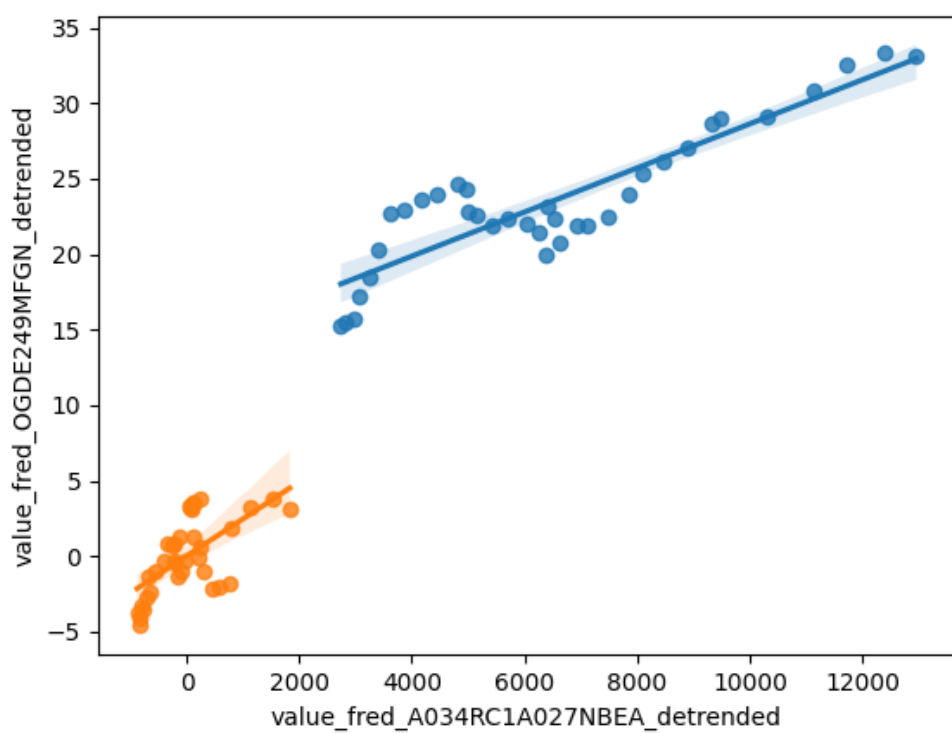
|                                             | coef       | std err | t        | P>  t | [0.025 | 0.975] |
|---------------------------------------------|------------|---------|----------|-------|--------|--------|
| <b>const</b>                                | -5.583e-14 | 0.328   | -1.7e-13 | 1.000 | -0.667 | 0.667  |
| <b>value_fred_A034RC1A027NBEA_detrended</b> | 0.0025     | 0.001   | 4.770    | 0.000 | 0.001  | 0.004  |

|                       |       |                          |       |
|-----------------------|-------|--------------------------|-------|
| <b>Omnibus:</b>       | 0.572 | <b>Durbin-Watson:</b>    | 0.205 |
| <b>Prob(Omnibus):</b> | 0.751 | <b>Jarque-Bera (JB):</b> | 0.649 |
| <b>Skew:</b>          | 0.067 | <b>Prob(JB):</b>         | 0.723 |
| <b>Kurtosis:</b>      | 2.356 | <b>Cond. No.</b>         | 638.  |

Notes:

[1] Standard Errors assume that the covariance matrix of the errors is correctly specified.



**Figure 10:** Detrended Regression Plot for 2026-09-11

## 7 Date: 2026-09-12

### Series ID: CUUSA200SAH1

This series is titled Consumer Price Index for All Urban Consumers: Shelter in Midwest - Size Class A and has a frequency of Semiannual. The units are Index 1982-1984=100 and the seasonal adjustment is Not Seasonally Adjusted. The observation start date is 1984-01-01 and the observation end date is 2026-01-01. The popularity of this series is 1.

### Series ID: APU0400706111

This series is titled Average Price: Chicken, Fresh, Whole (Cost per Pound/453.6 Grams) in the West Census Region - Urban and has a frequency of Monthly. The units are U.S. Dollars and the seasonal adjustment is Not Seasonally Adjusted. The observation start date is 1980-01-01 and the observation end date is 2026-08-01. The popularity of this series is 14.

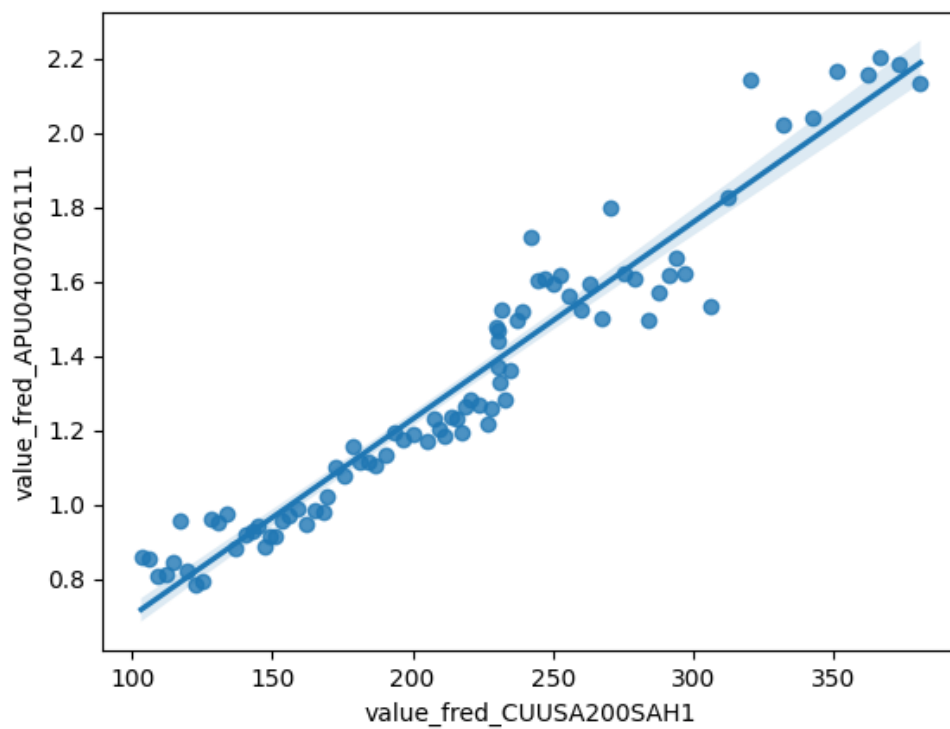
## 7.1 Raw Regression

Regresses the two series against each other directly. A significant result here can come just as easily from both series sharing a long-run trend as from any real relationship.

|                         |                          |                     |          |       |        |        |
|-------------------------|--------------------------|---------------------|----------|-------|--------|--------|
| Dep. Variable:          | value_fred_APU0400706111 | R-squared:          | 0.938    |       |        |        |
| Model:                  | OLS                      | Adj. R-squared:     | 0.937    |       |        |        |
| Method:                 | Least Squares            | F-statistic:        | 1248.    |       |        |        |
| Date:                   | Sat, 12 Sep 2026         | Prob (F-statistic): | 8.79e-52 |       |        |        |
| Time:                   | 14:28:28                 | Log-Likelihood:     | 79.771   |       |        |        |
| No. Observations:       | 85                       | AIC:                | -155.5   |       |        |        |
| Df Residuals:           | 83                       | BIC:                | -150.7   |       |        |        |
| Df Model:               | 1                        |                     |          |       |        |        |
| Covariance Type:        | nonrobust                |                     |          |       |        |        |
|                         | coef                     | std err             | t        | P>  t | [0.025 | 0.975] |
| const                   | 0.1732                   | 0.034               | 5.099    | 0.000 | 0.106  | 0.241  |
| value_fred_CUUSA200SAH1 | 0.0053                   | 0.000               | 35.326   | 0.000 | 0.005  | 0.006  |
| Omnibus:                | 3.941                    | Durbin-Watson:      | 0.840    |       |        |        |
| Prob(Omnibus):          | 0.139                    | Jarque-Bera (JB):   | 3.180    |       |        |        |
| Skew:                   | 0.421                    | Prob(JB):           | 0.204    |       |        |        |
| Kurtosis:               | 3.436                    | Cond. No.           | 742.     |       |        |        |

Notes:

[1] Standard Errors assume that the covariance matrix of the errors is correctly specified.



**Figure 11:** Raw Regression Plot for 2026-09-12

## 7.2 Detrended Regression

Regresses the residuals of each series after removing its own linear time trend, so a shared trend can no longer drive the result on its own. A relationship that stays significant here is better evidence of a genuine link between the two series.

|                          |                                    |                            |          |
|--------------------------|------------------------------------|----------------------------|----------|
| <b>Dep. Variable:</b>    | value_fred_APU0400706111_detrended | <b>R-squared:</b>          | 0.370    |
| <b>Model:</b>            | OLS                                | <b>Adj. R-squared:</b>     | 0.363    |
| <b>Method:</b>           | Least Squares                      | <b>F-statistic:</b>        | 48.77    |
| <b>Date:</b>             | Sat, 12 Sep 2026                   | <b>Prob (F-statistic):</b> | 6.57e-10 |
| <b>Time:</b>             | 14:28:28                           | <b>Log-Likelihood:</b>     | 79.794   |
| <b>No. Observations:</b> | 85                                 | <b>AIC:</b>                | -155.6   |
| <b>Df Residuals:</b>     | 83                                 | <b>BIC:</b>                | -150.7   |
| <b>Df Model:</b>         | 1                                  |                            |          |
| <b>Covariance Type:</b>  | nonrobust                          |                            |          |

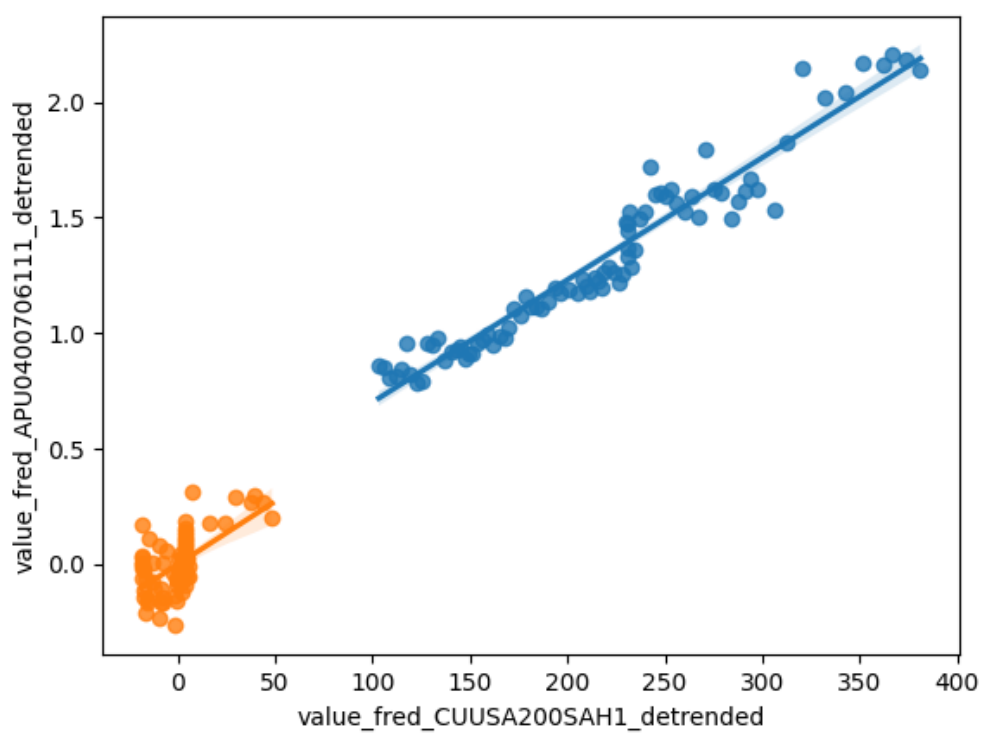
|                                          | coef      | std err | t        | P>  t | [0.025 | 0.975] |
|------------------------------------------|-----------|---------|----------|-------|--------|--------|
| <b>const</b>                             | 2.172e-14 | 0.010   | 2.09e-12 | 1.000 | -0.021 | 0.021  |
| <b>value_fred_CUUSA200SAH1_detrended</b> | 0.0055    | 0.001   | 6.984    | 0.000 | 0.004  | 0.007  |

|                       |       |                          |       |
|-----------------------|-------|--------------------------|-------|
| <b>Omnibus:</b>       | 4.280 | <b>Durbin-Watson:</b>    | 0.842 |
| <b>Prob(Omnibus):</b> | 0.118 | <b>Jarque-Bera (JB):</b> | 3.513 |
| <b>Skew:</b>          | 0.442 | <b>Prob(JB):</b>         | 0.173 |
| <b>Kurtosis:</b>      | 3.459 | <b>Cond. No.</b>         | 13.3  |

Notes:

[1] Standard Errors assume that the covariance matrix of the errors is correctly specified.



**Figure 12:** Detrended Regression Plot for 2026-09-12